PharmaCare + PharmaIQ — User Guide

How to add your data, what to type, and a tour of every page.

Written for pharmacy staff, pharmacists and managers — no programming needed.

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1. Getting started

1.1 Start the app

```
cd PharmaCare-PharmaIQ
python -m venv venv
source venv/bin/activate            # Windows: venv\Scripts\activate
pip install -r requirements.txt
python seed_ai_demo.py              # recommended: demo sales, fresh expiry dates, sample documents
python app.py
```

Open **http://127.0.0.1:5000**.

> **Why run `seed_ai_demo.py`?** It makes the demo *fresh*: it moves medicine expiry dates so some are expired and some expire soon, adds ~120 days of sales (needed for forecasts and anomaly detection), and loads 12 sample documents. Without it the AI still works, but expiry/sales questions may return "nothing found" because the original demo dates are old. It is safe to re-run; `python seed_ai_demo.py --reset` removes only the sales it created.

1.2 Log in

Role	Username	Password	Use it to see…
Admin	`admin`	`admin123`	everything
Pharmacist	`pharmacist`	`pharma123`	everything except audit, evaluation, document deletion
Staff	`staff`	`staff123`	the limited view — great for testing permissions

Open a second browser (or a private window) to compare roles side by side.

1.3 Find the AI features

After login, the left sidebar has a new section **PHARMAIQ · AI**:

Link	What it is	Who sees it
✨ **Ask PharmaIQ**	chat — ask in plain English	everyone
📖 **Knowledge Base**	leaflets & SOPs the AI can cite	everyone (upload: pharmacist/admin)
📈 **AI Insights**	forecast, anomalies, reorder list	admin, pharmacist
🛡 **AI Audit Trail**	log of every AI question	admin
✅ **AI Evaluation**	one-click quality scorecard	admin

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2. Adding your data (the inputs)

PharmaIQ answers from **three kinds of input**. Better input → better answers.

2.1 Medicines (inventory)

Sidebar → Medicines → "Add Medicine". (Admin / Pharmacist.)

Field	Required	What to enter	Why PharmaIQ cares
Medicine Name	✅	e.g. `Paracetamol 500mg`	the name PharmaIQ matches in questions (first word, e.g. *paracetamol*, is enough; small typos are tolerated)
Medicine Type	✅	`Tablet`, `Capsule`, `Syrup`, `Injection`, `Inhaler` …	this is the **category** — "show only tablets" uses it. Keep spelling consistent
Company Name		`Cipla Ltd`	"medicines from Cipla"
Reference No / Lot No		batch identifiers	lot appears as **batch** in expiry answers
Issue Date / **Expiry Date**	✅	`YYYY-MM-DD` date pickers	expiry questions and alerts; PharmaIQ flags expiry before issue date
Price	✅	number	stock value, "priced above $20"
Quantity	✅	whole number	stock, low-stock (≤ 10 by default)
Dosage / Uses / Side Effects / Precautions		free text	used only as a **fallback**, labelled "internal record — not an approved leaflet", when no document matches
Barcode		digits	duplicate barcodes are flagged as data-quality issues
Image		JPG/PNG ≤ 5 MB	not used by AI

Tips * One row = one **batch**. Two batches of the same medicine = two rows (PharmaIQ lists them separately). * Never leave the expiry date blank — blank expiry is reported as a data issue. * Bulk loading: `python seed_data.py` loads 30 sample medicines; the REST API (`POST /api/medicines`) can load more.

2.2 Sales

Sidebar → Sales → "New Sale" — choose the medicine, type a quantity (the page shows stock, unit price and total), save. The app reduces stock automatically and records who sold it.

- PharmaIQ's sales answers, forecasts and reorder suggestions all come from these records.
- **Forecasts need history:** at least **14 days** of sales; 28+ days is better (weekday patterns), 56+ is best. `seed_ai_demo.py` provides 120 days.
- Each person's sales are tagged with their username — **staff users can only ask about their own sales.**

2.3 Documents (leaflets & SOPs) — the knowledge input

Sidebar → Knowledge Base → "Upload a document" (Pharmacist / Admin).

Setting	Options
File	`.pdf`, `.docx`, `.txt`, `.md` — up to **5 MB**
Title	optional (auto-detected from the first heading)
Type	Leaflet (medicine information) · SOP (procedure/policy) · Reference
Visible to	**All staff** · **Pharmacists+** · **Admins only** (admin-only documents are invisible to everyone else, even in AI answers)

What makes a document work well: * **Use real text.** Scanned PDFs (photos of paper) are rejected — run OCR first. * **Use headings.** Word "Heading 1/2" styles or Markdown `#` headings become the *section* shown in citations. * **One topic per document** (one leaflet per medicine; one SOP per procedure) — citations are then precise. * **Put the medicine name in the title** ("Metformin 500 mg Tablets — Product Information"). Questions that name a medicine are matched to that leaflet. * Upload only **approved, current** material. PharmaIQ quotes it; it does not check it clinically. * Uploading the same file twice is detected and ignored.

Try it — save this as `visitor_policy.md` and upload as *SOP / All staff*:

```
# SOP-07 Visitor Policy

## Reception
All visitors must sign the register at reception and wear a visitor badge.

## Restricted areas
Visitors are never allowed in the dispensary or the controlled drugs cabinet area.
```

Then ask PharmaIQ: **"Where must visitors sign in?"** — you should get the sentence with a numbered source *SOP-07 Visitor Policy*.

Sample documents — 12 demo files live in the `knowledge_base/` folder (6 leaflets, 6 SOPs: returns, expired stock, cold chain, reorder, controlled medicines, counselling scope). Load them with **Knowledge Base → "Load sample documents"** (admin) or `python seed_ai_demo.py`. They are *demonstration content*, not clinical references.

2.4 The questions (your main input)

Type naturally — full sentences, short phrases, or keywords all work. Names don't need exact spelling. You can include a number (`next 45 days`, `fewer than 20 units`), a category (`tablets`), a company (`Cipla`), a period (`this month`, `last 7 days`, `yesterday`) or a date (`expiring before 2027-01-31`). See section 4.

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3. Website tour — page by page

3.1 Original PharmaCare pages

Page	What you do there
Dashboard	totals, stock value, 7-day sales chart, alerts at a glance
Medicines	search, add / edit / delete, view details, scan barcode, pagination
Barcode Scanner	scan with the camera to find a medicine instantly
Alerts	three lists: expiring ≤ 30 days · already expired · low stock ≤ 10
Sales	record a sale, see sales history
Reports *(admin, pharmacist)*	totals by type, expiry list, 30-day sales summary
Users / Activity Logs *(admin)*	manage accounts; see who did what
Export Excel	colour-coded inventory workbook (red = expired, yellow = low stock)
Profile	change your own password

PharmaIQ uses the **same rules** as the Alerts page (low stock ≤ 10, 30-day expiry window), so the AI and the page agree.

3.2 ✨ Ask PharmaIQ

- **Chat box** at the bottom: type and press Enter.
- **"Try asking" chips** on the right: click one to ask it instantly.
- **Mode badge** (top right) shows how questions are planned: `offline (rule-based)` by default, or `anthropic:…` / `openai-compatible:…` if an LLM is configured.
- **New chat** clears the conversation (follow-ups start fresh).

Each answer can contain:

Block	Colour	Meaning
Database facts · as of DATE	green	numbers computed from your real data, a chart, and the result table
Show query	grey button	the exact SQL that produced the table — nothing hidden
Excel / CSV	green buttons	download the result (the Excel file also records the question, date and SQL)
From documents + **Sources**	blue	text taken from leaflets/SOPs with numbered citations `[1]`, `[2]`
Explanation	purple	short commentary, labelled *rule-based* or *AI-generated* — it is never the source of the numbers
Warnings / assumptions	amber / grey	defaults that were applied (e.g. "assumed last 30 days"), caveats such as "forecasts are estimates"
Follow-up chips	pills	click to continue the conversation

3.3 📖 Knowledge Base

- Table of documents: title, type, who can see it, number of searchable passages, date.
- **Upload** (pharmacist/admin), **Load sample documents** and **delete** (admin).

- **Retrieval backend** line at the top (`tfidf` by default). If an optional upgrade failed to load, an amber note explains why.
- **Test retrieval** box: type a query and see which passages match and *why* (keyword score, vector score, coverage; ✔ = counted as relevant, ✖ = too weak to cite). Use it to check whether a new document is findable.

3.4 📈 AI Insights *(admin, pharmacist)*

Panel	How to use	How to read it
Demand forecast	pick a medicine (or "All") and 7 / 14 / 30 days → *Forecast*	grey line = actual last 30 days; green = forecast; dashed = **80 % range**. Facts list the expected total, confidence (low/medium/high), a 7-day back-test, and days of cover for one medicine. **A wide band = low certainty.** If history is too short it says so instead of guessing
Anomalies & data quality	automatic; *Refresh*	`sales_spike` / `sales_drop` (unusually high/low days) and data issues: `sale_total_mismatch`, `sold_after_expiry`, `duplicate_barcode`, `negative_stock`, `expiry_before_issue`, `missing_expiry`, `sale_unknown_medicine`. Severity high/medium/low. These are **flags to check**, not accusations
Reorder suggestions	automatic	items at/below the low-stock level or with ≤ 14 days of cover (from the last 30 days' sales), with a *suggested order quantity* — rules are printed under the table. A pharmacist should confirm before ordering

With demo data you should see: one **sales_spike** for the medicine with the injected 70-unit day, and one **sale_total_mismatch** (a sale whose total doesn't equal quantity × price). Those two are deliberate.

3.5 🛡 AI Audit Trail *(admin)*

Every AI request: time, user and role, the (privacy-redacted) question, route (`tool` / `sql` / `rag` / `refused` …), tool name, expandable SQL, row count, status (`ok`, `refused`, `denied`, `error`, `no_answer`) and latency. Filter by status. It stores **what was asked and how it was answered — never the returned rows** — and strips e-mail addresses and phone-like numbers from questions.

3.6 ✅ AI Evaluation *(admin)*

Click **Run evaluation** (a few seconds). You get a scorecard: * **SQL correctness** — each test question is compared with an independent control query on your database. * **RAG pass rate, hit@1, hit@3, keyword recall** — does the right document come first? * **Safety** — must-refuse questions, must-not-refuse questions, and SQL attack attempts.

A healthy system shows 100 %. If you upload new documents or change data and a score drops, the failing row tells you which case to look at. You can add your own cases in `evaluation/*.json`.

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4. Ask PharmaIQ — question cookbook

"Route" tells you where the answer comes from. **DB** = live database, **Doc** = documents, **Calc** = statistics.

4.1 Expiry

Type this	Route	You get
Which medicines are expiring within the next 30 days, and what is their current stock?	DB tool	name, batch, expiry date, quantity, days left, stock value; total units and value at risk
What expires in the next 90 days?	DB tool	same, 90-day window
Medicines expiring this month	DB tool	days to month end
Show already expired medicines	DB tool	expired batches + days overdue + a reminder to follow the SOP
Tablets expiring in the next 60 days	DB tool	filtered by category
Medicines expiring before 2027-03-31	DB (text-to-SQL)	custom date filter (also lists batches that already expired — the assumptions line says so)

4.2 Stock

Type this	Route	You get
Show low stock items	DB tool	items ≤ 10 units (rule shown)
Show all products with fewer than 10 units	DB (text-to-SQL)	strict “< 10” — compare with the line above
What is out of stock?	DB tool	threshold 0
How much stock of amoxicillin do we have?	DB tool	per batch + status (OK / Low / Expiring soon / Expired)
How many tablets do we have?	DB (text-to-SQL)	a count
Medicines priced above $20	DB (text-to-SQL)	list
Most expensive capsules	DB (text-to-SQL)	sorted list
Total value by category · inventory value by company	DB tool	stock value table + chart
Medicines from Cipla with less than 50 units	DB (text-to-SQL)	combined filters

4.3 Sales

Type this	Route	You get
Which categories generated the highest sales this month?	DB tool	revenue per category + bar chart
Top 5 best selling medicines last month	DB tool	units, revenue
Sales by day last 7 days	DB tool	line chart
Total sales this year · sales today	DB tool	transactions, units, revenue
Sales of paracetamol last month	DB (text-to-SQL)	daily units and revenue
Who sold the most this month?	DB tool	per seller *(staff see only themselves)*
Give me a management summary	DB tool	5 tables: inventory KPIs, low stock, expiring, sales by category, top sellers — downloadable as a multi-sheet Excel report

4.4 Intelligence *(admin / pharmacist)*

Type this	Route	You get
Forecast demand for the next 14 days	Calc	expected units with 80 % range, confidence, back-test
Forecast demand for metformin next 2 weeks	Calc	plus days of cover for that medicine (needs recent sales for that item — a medicine that stopped selling forecasts ≈ 0)
Any unusual sales spikes? / Find anomalies	Calc + DB	spikes, drops, data-quality findings
What should we reorder?	Calc + DB	days of cover and suggested quantities

4.5 Documents

Type this	Route	You get (with sample documents)
What is the return policy?	Doc	SOP-01 passages, cited
How should insulin be stored?	Doc	refrigerate 2–8 °C, do not freeze, in-use pen ≤ 28 days — from the insulin leaflet
What are the side effects of metformin?	Doc	nausea, diarrhoea… from the metformin leaflet only
What is the procedure for expired stock?	Doc	quarantine, register, disposal (SOP-02)
What temperature should the pharmacy fridge be?	Doc	SOP-03 (“fridge” = “refrigerator”)
What are the warnings for warfarin?	Doc	bleeding, INR monitoring, interactions
Who can give advice on which medicine to take?	Doc	SOP-06: only a registered pharmacist
paracetmol overdose liver	Doc	typo tolerated → paracetamol leaflet

If nothing relevant is found you get an honest *“I couldn’t find this… I won’t guess”* instead of an invented answer.

4.6 Questions it should decline

See section 7.

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5. How to read an answer

1. **Check the “as of” date** next to *Database facts*. It is today’s date, taken at the moment you asked.
2. **Read the assumptions** (grey bullets): the period used, the low-stock rule, the forecast model.
3. **Look at the table and, if in doubt, click Show query.** The table is real data; the SQL shows exactly what was asked of the database.
4. **Treat the purple Explanation as a hint.** It may be rule-based or AI-written; it is checked so it cannot contain numbers that aren’t in the data, but it is not evidence.
5. **For document answers, check the Sources** — the bold title, section and page tell you where the sentence came from. Open the original document for anything important.
6. **For forecasts, read the range and the confidence**, not just the headline number.
7. **Anything clinically consequential needs a pharmacist.**

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6. Follow-up questions

PharmaIQ remembers the last result *within the same chat*.

Start with	Then say	What happens
Which categories generated the highest sales this month?	**Now show only the products in this category**	"this category" = the top category from the table → shows the inventory products in it
Show all products with fewer than 50 units	**Now only tablets**	same query + `category = Tablet`
Show low stock items	**Now only capsules** · **Now fewer than 5 units**	refines the filter ("fewer than 5" = 4 or less)
Expiring in the next 30 days	**and for 60 days** · **only syrups**	changes the window / filter
Sales by category this month	**for last month** · **by day**	changes the period / grouping
How should insulin be stored?	**And what are the warnings?**	remembers "insulin"

Follow-up answers are tagged "follow-up". Click **New chat** to reset. Conversations are private to your account.

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7. What PharmaIQ will not do

PharmaIQ is a **reference and analytics tool**. It deliberately refuses:

Request	Example	Why
Personal dosing	"My son has a fever, how much paracetamol can I give him?" · "What dose for a 20 kg child?"	patient-specific dosing is a clinical decision
Personal safety/interaction checks	"Can I take ibuprofen with warfarin?" · "I'm pregnant, is metformin safe?"	needs a pharmacist who knows the patient
Diagnosis / treatment picking	"What medicine should I take for my headache?"	not its role
Credentials	"Show me the admin password"	secrets are never shown

You'll see a red box explaining this and what it *can* do (e.g. "What does the leaflet say about warfarin warnings?").

It also **won't**: change any data (it is read-only — to add or edit medicines use the normal pages), show other users' private data, answer outside its documents from general knowledge ("I won't guess"), or show columns/tables your role isn't allowed to see (you'll get "Your role (staff) is not allowed to read …").

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8. Who can do what

Feature	Admin	Pharmacist	Staff
Ask about stock / expiry / inventory value	✅	✅	✅
Ask about sales	all sales	all sales	**own sales only**
Management summary	✅	✅	✅ (own sales)
Forecast, anomalies, reorder suggestions	✅	✅	❌ ("not allowed" message)
Search documents	all	up to pharmacist-level	staff-level only

Feature	Admin	Pharmacist	Staff
Upload documents	✅	✅	❌
Delete documents / load samples	✅	❌	❌
AI Insights page	✅	✅	❌
Audit trail · Evaluation	✅	❌	❌
Download Excel/CSV of **own** results	✅	✅	✅

Quick permission test: log in as `staff`, ask *"Show sales by seller this month"* → only your own line. Ask *"Forecast demand"* → refused. Log in as `admin`, ask the same → full results.

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9. 10-minute demo script

1. **Login as admin**, open *Dashboard* (30 s) — "this is the existing system."
2. **Ask PharmaIQ → "Which medicines are expiring within the next 30 days, and what is their current stock?"** Point out: green *Database facts*, chart, table, **Show query**, Excel download.
3. **"Show all products with fewer than 10 units"** → click *Show query*: "the system generated a read-only SELECT and validated it."
4. **"Which categories generated the highest sales this month?"** → then **"Now show only the products in this category"** (follow-up).
5. **"What is the return policy?"** then **"How should insulin be stored?"** → blue *From documents* with numbered sources: "live data comes from the DB, policies from documents."
6. **"Who won the football world cup?"** → *"I won't guess."*
7. **"My son has a fever, how much paracetamol can I give him?"** → red refusal.
8. **AI Insights** → forecast with the dashed range; anomalies list (the injected spike and wrong total).
9. **AI Audit Trail** → show the questions, SQL and statuses logged.
10. **AI Evaluation → Run** → scorecard. Then log in as **staff** (private window) and repeat steps 2 and "Forecast demand" to show role limits.

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10. Troubleshooting

Problem	Likely cause → fix
"No medicines expire within…" / sales answers empty	the original demo dates are old → run `python seed_ai_demo.py`
Knowledge Base is empty / document questions say "I couldn't find this"	click **Load sample documents** (admin) or upload documents
Uploaded PDF rejected: "No selectable text found"	it is a scan → OCR it, or upload a DOCX/TXT/MD version
Uploaded file says "already in the knowledge base"	identical file was uploaded before
My document isn't cited	use **Test retrieval** with your question; add the medicine/topic name to the **title**, use headings, keep one topic per document
"Security token missing or expired"	reload the page (session expired) and try again
"Your role (staff) is not allowed to read …"	permission limit by design — ask an admin/pharmacist
Forecast says "Not enough sales history"	needs ≥ 14 days of sales for that medicine; try "All medicines" or add data

Problem	Likely cause → fix
Question answered by documents when you wanted data (or vice-versa)	rephrase with the data words (“how many”, “stock”, “expiring”, “sales”) or the document words (“policy”, “procedure”, “side effects”, “stored”)
Mode badge says *offline (rule-based)*	normal. To use an LLM, set `PHARMAIQ_LLM_PROVIDER` and an API key (see `.env.example`) and restart
Amber banner “sentence-transformers unavailable”	optional upgrade not installed; the app fell back to the built-in search — nothing is broken
Page not found / blank after update	restart `python app.py`; the new tables are created automatically on start

Need more detail? See **TECHNICAL_GUIDE.md**.

Sample leaflets and SOPs are demonstration content. PharmaIQ provides reference information and data analytics, not medical advice.